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Topological Kinematics

1 message

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Dear Pola and Hugo,

I have been thinking about a new TDA idea that I would really like to explore with you. It is still at a very early stage but I think there may be something interesting here. For now I am calling the idea

Topological Kinematics.

The basic idea: Suppose we have temporal data and analyse it using consecutive or overlapping time windows. For each window, we construct the appropriate point cloud/filtration and compute a persistence diagram. We therefore obtain a sequence

$$D_0, D_1, \dots, D_T.$$

Normally, we analyse these persistence diagrams individually, extract scalar summaries from them, or compare selected diagrams using bottleneck/Wasserstein distances.

My idea is to change the viewpoint slightly. Instead of thinking of $D_0, D_1, \dots, D_T$ simply as a collection of persistence diagrams, we regard them as a trajectory through the metric space of persistence diagrams: $t \mapsto D_t$.

Then we can ask questions analogous to those asked in ordinary kinematics.

For example:

- Where is the system topologically at a particular time? à **position**
- How quickly is its topology changing? à **speed/velocity**
- Is that rate of change itself increasing or decreasing? à **acceleration**
- Is the topological evolution continuing in roughly the same direction, or changing direction? à **turning**
- How much topological change has accumulated during the whole period? à **path length**
- How different is the final state from the initial state? à **net displacement**
- Has the system moved steadily from one structure to another, or undergone substantial restructuring and eventually returned close to where it started? à **trajectory efficiency / instability**

A simple analogy is to think about taking photographs of a moving person. A single photograph tells us where the person is. A sequence of photographs lets us study the motion itself: speed, acceleration, direction, distance travelled, etc.

Persistence diagrams would play the role of the photographs of the shape of the data, while Topological Kinematics would study the motion between them.

Literature Review: I have been looking into the literature because obviously the first question is how much of this has already been done. There is important related work:

- Cohen-Steiner, Edelsbrunner & Morozov (2006), “[Vines and Vineyards by Updating Persistence in Linear Time](#)”.**
 They introduced persistence vineyards, where persistence diagrams vary with a parameter and individual persistence points can be followed through time as “vines”. This is probably the earliest major framework explicitly concerned with evolving persistence information. The emphasis, however, is on tracking persistent features rather than developing kinematic descriptors of the entire diagram-valued trajectory.
- Kramár et al. (2016), “[Analysis of Kolmogorov Flow and Rayleigh–Bénard Convection Using Persistent Homology](#)”.**
 They compute persistence diagrams along time series of dynamical systems and use the metric structure of persistence-diagram space to study temporal dynamics. This is an important precursor because it already shows that the distances between diagrams contain information about how a system evolves.
- Xian, Adams, Topaz & Ziegelmeier (2022), “[Capturing Dynamics of Time-Varying Data via Topology](#)”.**
 This paper gives a broader framework for TDA of time-varying systems and discusses vineyards, crocker plots, multiparameter approaches and crocker stacks. The focus is on topological summaries of dynamic data, rather than developing a kinematic calculus for trajectories of persistence diagrams.
- Giusti & Lee (2023), “[Signatures, Lipschitz-Free Spaces, and Paths of Persistence Diagrams](#)”.**
 This is particularly important for us. They explicitly regard sequences of persistence diagrams as **paths of persistence diagrams**. They embed persistence-diagram space into a Lipschitz-free Banach space and use path signatures to characterize dynamic behaviour. Thus, the idea of a persistence-diagram trajectory itself is definitely not new. Their aim, however, is to construct representations/signatures of these paths rather than quantities analogous to speed, acceleration, turning, displacement, etc.
- Khormali (2025), “[Hierarchical Persistence Velocity for Network Anomaly Detection](#)”.**
 This introduces a velocity-based topological descriptor for dynamic networks and proves stability results. The construction measures the rate at which persistent features appear/disappear and is therefore another important indication that “velocity” language is already entering TDA. It is nevertheless different from treating the complete persistence diagram as a moving state in a metric space.
- Malhotra et al. (2026), “[Tracking Representation Dynamics in Large Language Models with Persistent Homology](#)”.**
 They study persistence diagrams across training checkpoints of large language models and quantify topological change between consecutive checkpoints using Wasserstein distances. They explicitly interpret this temporal evolution as topological activity/velocity. Therefore, simply defining consecutive Wasserstein distance as “topological velocity” would not be a novel contribution by itself.
- Bernal-Alvarado, Delepine & Pinedo Guadarrama (2026), “[Structural Divergence of the Roman–Byzantine Trade Network, 0–1453 CE: Persistent Homology, Topological Velocity, and Criticality Indicators of Imperial Collapse](#)”.**
 This very recent work explicitly defines a Wasserstein/topological velocity of the form $\dot{W}(t) = \frac{w_2(D_t, D_{t+\Delta t})}{\Delta t}$. Again, this confirms that scalar topological velocity based on consecutive persistence diagrams already exists.

So where could the novelty be?

After seeing this literature, I do **not** think we should claim that the novelty is simply to regard persistence diagrams as a trajectory, nor that $W(D_t, D_{t+1})$ is a new notion of topological velocity. Both ideas have clear precedents. What I have *not* found so far is a **systematic, domain-independent kinematic framework for persistence-diagram trajectories**. In other words, rather than defining one velocity statistic for a particular application, the idea would be to develop something more like

temporal data $\rightarrow$ persistence-diagram trajectory $\rightarrow$ $\left\{ \begin{array}{l} \text{speed,} \\ \text{acceleration,} \\ \text{direction/turning,} \\ \text{path length,} \\ \text{net displacement,} \\ \text{trajectory efficiency,} \\ \text{structural instability.} \end{array} \right.$

The distinction between **path length** and **net displacement** is particularly simple but potentially useful:

A system could begin at D_0 , undergo enormous structural reorganization, and eventually return close to its original state D_T . We could then have $d(D_0, D_T) \approx 0$ while $\sum_{t=0}^{T-1} d(D_t, D_{t+1})$ is very large. Static comparison of the beginning and end would say “almost nothing changed”. The trajectory says “a lot happened, but the system eventually came back”.

Even more interesting is **direction and acceleration**. Persistence-diagram space is not an ordinary Euclidean vector space, so we cannot simply write $D_{t+1} - D_t$. Therefore, developing meaningful notions of changing direction and acceleration is actually a mathematical problem rather than just terminology.

A possible step-by-step framework

Here is roughly how I currently imagine the pipeline.

1. Start with temporal data.

Suppose we observe a time-dependent system $X(t)$.

This could be a financial time series, dynamical system, network, biological system, etc.

2. Construct temporal windows.

Choose windows $W_0, W_1, \dots, W_T$.

These may overlap. The window size and step size will obviously influence the temporal resolution and will need to be studied carefully.

3. Construct a topological object for every window.

For each W_t , build the appropriate object X_t .

For a time series, for example, this may involve a time-delay embedding inside each window: $W_t \rightarrow X_t$.

The sliding window and the time-delay embedding should be regarded as two different steps.

4. Compute persistent homology.

For each X_t , compute persistence diagrams $D_t^{(0)}, D_t^{(1)}, \dots$ for the homological dimensions of interest. Then, for each k , $\gamma_k: t \mapsto D_t^{(k)}$ becomes a discrete trajectory in persistence-diagram space.

5. Define local topological speed.

Choose a metric d , for example W_p or bottleneck distance, and define $v_t = \frac{d(D_t, D_{t+1})}{t_{t+1} - t_t}$.

This tells us how rapidly the topology changes locally. For equally spaced windows this is essentially the consecutive diagram distance up to a constant.

6. Define cumulative path length.

Over an interval, $L = \sum_{t=0}^{T-1} d(D_t, D_{t+1})$. This gives the total amount of topological motion experienced by the system.

7. Define net topological displacement.

$$R = d(D_0, D_T).$$

This measures how different the final topological state is from the initial state.

8. Compare path length with displacement.

For example, $\eta = \frac{R}{L}$, $0 \leq \eta \leq 1$, when $L > 0$. This could provide a simple measure of how “direct” the topological evolution was.

If $\eta \approx 1$, most of the motion contributed to moving the system away from its initial state.

If $\eta \ll 1$, the system travelled considerably through diagram space but ultimately returned near its starting state.

We need to think carefully about the name (perhaps topological straightness, trajectory efficiency, or something similar).

9. Measure changes in speed.

A first scalar analogue of acceleration could be $a_t^{\text{speed}} = \frac{v_t - v_{t-1}}{\Delta t}$.

Large positive values would correspond to rapidly increasing structural change; negative values would indicate stabilization. But this cannot be the whole definition of acceleration.

10. Introduce direction/turning.

This is probably where the mathematics becomes much more interesting.

Take three consecutive persistence diagrams D_{t-1}, D_t, D_{t+1} . Let

$a = d(D_{t-1}, D_t)$, $b = d(D_t, D_{t+1})$, $c = d(D_{t-1}, D_{t+1})$. These three distances form a metric triangle. One

possibility is to define a comparison angle through $\cos \theta_t = \frac{a^2 + b^2 - c^2}{2ab}$. Then θ_t tells us whether the trajectory approximately continues in the same direction or bends sharply in diagram space.

This matters because a trajectory could move at constant speed while continuously changing direction. A scalar derivative of speed would give zero acceleration, although the trajectory is clearly not dynamically

stationary. There may also be a deeper approach using the geometry of Wasserstein persistence-diagram space, optimal matchings, geodesics and tangent cones. I think we should investigate this carefully before fixing the definition.

11. Construct local instability descriptors.

Once we have the speed sequence $v_0, v_1, \dots, v_{T-1}$, we can define local quantities measuring fluctuations in speed and/or turning over a neighbourhood of t .

The goal would be to distinguish:

- stable topological evolution,
- rapid transition,
- sustained directional change,
- oscillatory/reversible restructuring,
- sudden structural shocks.

This part needs much more thought because I do not want to create arbitrary indices just for the sake of having more descriptors.

12. Establish stability.

We should prove that these descriptors behave continuously when the underlying persistence diagrams are perturbed.

For speed, for example, if $d(D_t, \tilde{D}_t) \leq \varepsilon_t$, then the triangle inequality gives

$$|d(D_t, D_{t+1}) - d(\tilde{D}_t, \tilde{D}_{t+1})| \leq \varepsilon_t + \varepsilon_{t+1}.$$

Therefore, $|v_t - \tilde{v}_t| \leq \frac{\varepsilon_t + \varepsilon_{t+1}}{\Delta t}$.

So at least some stability results should follow quite naturally from the metric structure.

The more interesting question will be stability of the directional/turning and acceleration quantities.

13. Test the framework first on synthetic examples.

I think synthetic experiments will be crucial because we can construct examples where we know exactly what is happening.

For example:

- constant-speed evolution;
- accelerating evolution;
- a sudden structural transition;
- a trajectory that moves away and then returns;
- two trajectories having the same sequence of speeds but different directions;
- two trajectories with the same initial and final persistence diagrams but completely different intermediate histories.

If our descriptors distinguish these cases in the expected way, that would make the interpretation very transparent.

14. Apply it to real temporal data.

Only after validating the framework synthetically would we use one or more real datasets.

Here we can compare Topological Kinematics against standard persistence summaries, consecutive Wasserstein distances, persistence entropy, landscapes, etc., and ask whether the additional kinematic information identifies transitions or periods of instability that the existing summaries miss.

So, in short, I think the potentially novel idea is not “persistence diagrams change through time”. That is well established. It is to ask whether we can build a coherent **kinematics of persistence-diagram trajectories**, with mathematically meaningful notions of motion, acceleration, turning, accumulated displacement and instability, and to understand what additional information these quantities provide about temporal systems.

Let me know what you think.

Best,
Ioannis



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